

PSYCH-BENCH: A Comprehensive Cognitive Battery for Evaluating Holographic Liquid Brain Architectures

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Abstract

We present PSYCH-BENCH, a psychological benchmark suite comprising 84 benchmarks across 17 cognitive domains for evaluating Holographic Liquid Brain (HLB) architectures—systems combining Hyperdimensional Computing, Liquid Time-Constant networks, Integrated Information Theory, and Active Inference. Existing AI evaluation frameworks lack psychometric infrastructure (HELM), cover narrow domains (ARC), omit normative human comparison (BIG-Bench), or ignore neuromodulatory effects (CogBench). PSYCH-BENCH addresses all four gaps with: (1) implementations of established paradigms (Stroop, WCST, N-back, and 75 others) with psychometric assessment infrastructure including ICC reliability, speed-accuracy tradeoff curves, and adaptive staircases; (2) normative anchoring against 90+ published human baselines enabling z-score and clinical-band interpretation; (3) a 17-benchmark neuromodulator evaluation framework testing dose-response curves, tolerance/withdrawal dynamics, behavioral knockouts, and multi-transmitter synergy; and (4) a 14-indicator consciousness evaluation derived from six theories. All stimuli share a unified holographic encoding via five HDC adapter types, enabling cross-modal comparison. Demonstration on the SYMTHAEA HLB reveals characteristic strengths in similarity-based discrimination (Social, Executive, Reasoning) and weaknesses in sequential processing (Language), with the psychometric analysis exposing important architectural properties: a gradient of reliability from Excellent to Poor (median ICC = 0.23, with 5 benchmarks reaching Moderate or better), selective sensitivity to working-memory and encoding-noise ablation, and accuracy-pressure tradeoffs in 4 of 8 tested benchmarks. Implemented in 15,000 lines of Rust with 793 unit tests, PSYCH-BENCH is open-source and fully deterministic.

Keywords: cognitive benchmark, holographic computing, liquid neural networks, neuromodulation, 26
psychometrics, consciousness evaluation, hyperdimensional computing 27

1 Introduction 28

A new class of cognitive architecture has emerged at the intersection of hyperdimensional computing, 29
dynamical systems neuroscience, and consciousness science. These *Holographic Liquid Brain* (HLB) 30
architectures encode information as high-dimensional distributed representations—the “holographic” 31
component, drawing on Plate’s holographic reduced representations (Plate, 1995) and Kanerva’s 32
hyperdimensional computing framework (Kanerva, 2009)—while processing temporal dynamics 33
through liquid-state computation, specifically Liquid Time-Constant (LTC) networks (Hasani et al., 34
2021) and their closed-form continuous-time variant (CfC) (Hasani et al., 2022). The “liquid” 35
component provides biologically plausible temporal processing with continuous-time dynamics and 36
adaptive time constants, while integration with Integrated Information Theory (IIT) (Tononi, 2004; 37
Oizumi et al., 2014) and Active Inference under the Free Energy Principle (Friston, 2010; Friston 38
et al., 2017) creates architectures with rich internal dynamics that demand correspondingly rich 39
evaluation. 40

The fundamental challenge is methodological: how should we evaluate cognitive architectures 41
that aspire to neuroscience-grade fidelity? Standard AI benchmarks fall short in four critical ways. 42

Lack of psychometric rigor. The Holistic Evaluation of Language Models (HELM) (Liang 43
et al., 2023) evaluates across many tasks but treats each as an isolated accuracy measure without 44
psychometric validation—no test-retest reliability, no construct validity analysis, no speed-accuracy 45
decomposition. A benchmark that fluctuates wildly across runs tells us nothing stable about the 46
architecture it evaluates. 47

Narrow domain coverage. The Abstraction and Reasoning Corpus (ARC) (Chollet, 2019) 48
provides an excellent measure of fluid intelligence but covers a single cognitive domain. Human 49
cognition is characterized by the interplay between executive function, working memory, social 50
cognition, motor learning, and dozens of other capacities. Single-domain benchmarks cannot capture 51
the profile that emerges from an integrated architecture. 52

Absence of normative comparison. BIG-Bench (Srivastava et al., 2023) offers hundreds of tasks but provides no systematic comparison to human performance distributions. Without normative anchoring—z-scores against published baselines with known means and standard deviations—we cannot interpret whether an architecture’s performance on a given task is exceptional, average, or impaired relative to the human population it models.

The neuromodulator gap. Perhaps most critically, no existing benchmark suite tests the effects of pharmacological manipulations on cognitive performance. CogBench (Coda-Forno et al., 2024) applies eight cognitive psychology paradigms to large language models, representing the closest existing work, but its eight tasks and prompt-based interface cannot probe neuromodulatory dynamics. For HLB architectures with explicit dopaminergic, noradrenergic, serotonergic, and cholinergic subsystems, evaluating how these transmitters modulate behavior is essential—analogous to how pharmacological challenge studies validate computational models of the basal ganglia (Doya, 2002) or prefrontal cortex (Arnsten, 2011).

PSYCH-BENCH addresses all four gaps simultaneously. We present a comprehensive cognitive battery of 84 benchmarks spanning 17 domains, designed specifically for evaluating HLB and related neuroscience-inspired architectures. The suite is organized around five design principles: psychometric fidelity to source paradigms, unified holographic stimulus encoding, normative anchoring against published baselines, systematic ablation transparency, and self-contained reporting.

Our contributions are:

1. An 84-benchmark cognitive battery covering 8 profile domains (executive function, memory, attention, social cognition, motor learning, language, metacognition, reasoning) plus specialized sub-batteries for affect, creativity, inhibition, and consciousness.
2. A 17-benchmark neuromodulator evaluation framework—the first to test pharmacological manipulations (dose-response curves, injection challenges, tolerance/withdrawal, antagonist profiles, multi-transmitter synergy) in an AI benchmark context.
3. A psychometric infrastructure providing ICC(3,1) reliability, ex-Gaussian RT decomposition (Heathcote, 1996), Wickelgren SAT curves (Wickelgren, 1977), adaptive staircases (Levitt, 1971), and normative z-score comparison with clinical interpretation bands (Cicchetti, 1994).
4. A 14-indicator consciousness evaluation derived from Butlin et al.’s (Butlin et al., 2023) six-

theory framework, with ablation matrices linking architectural features to indicator presence. 82

2 Design Principles 83

PSYCH-BENCH is organized around five principles that distinguish it from existing AI evaluation 84
suites. 85

Psychometric fidelity. Each benchmark implements the exact experimental paradigm described 86
in its source publication. The Stroop task (Stroop, 1935) presents congruent and incongruent 87
color-word stimuli following MacLeod’s (MacLeod, 1991) standardized procedure. The Wisconsin 88
Card Sorting Test (Berg, 1948) implements 6-category set-shifting with perseverative error tracking. 89
The Iowa Gambling Task (Bechara et al., 1994) uses the original reward/punishment schedule across 90
100 trials with four decks. This is not approximation—each benchmark carries a citation to its 91
source paradigm, and the implementation can be audited against the original procedure. 92

Holographic stimulus encoding. All stimuli pass through a unified Hyperdimensional Com- 93
puting (HDC) encoding pipeline before reaching the cognitive architecture. Five adapter types— 94
SequenceAdapter, SemanticAdapter, SpatialAdapter, ScenarioAdapter, and RewardAdapter— 95
convert raw stimuli (token sequences, spatial coordinates, reward outcomes, scenarios) into continuous 96
hypervectors in $\mathbb{R}^D$ using binding ($\otimes$) and bundling ($\oplus$) operations (Kleyko et al., 2023). This 97
unified representation is the “holographic” in HLB: all modalities share a common geometric space 98
where similarity is cosine distance, enabling cross-modal retrieval and comparison. The default di- 99
mension $D = 512$ balances representational capacity against computational cost for benchmark-scale 100
evaluation. 101

Normative anchoring. Every benchmark has at least one associated human baseline from 102
the published literature, stored as a mean $\pm$ SD pair with citation. The BaselineCollection 103
contains 90+ such entries spanning all 17 domains. Agent performance is converted to z-scores 104
($z = (x_{\text{agent}} - \mu_{\text{human}})/\sigma_{\text{human}}$, negated for error metrics where lower is better) and mapped to 105
percentiles via the normal cumulative distribution function. Clinical interpretation bands follow 106
Cicchetti’s (Cicchetti, 1994) guidelines: Superior ($z > 2.0$), Above Average ($1.0 < z \leq 2.0$), Average 107

($-1.0 \leq z \leq 1.0$), Below Average ($-2.0 \leq z < -1.0$), and Impaired ($z < -2.0$).

Ablation transparency. Six predefined ablation presets systematically disable architectural subsystems: `FullConsciousness` (all systems active), `CfcOnly` (liquid dynamics only, no active inference or social cognition), `NoFep` (Free Energy Principle disabled), `NoSocial` (Theory of Mind disabled), `ReducedWm` (working memory capacity reduced from 7 to 3), and `HdcOnly` (minimal holographic encoding). Each preset produces a complete cognitive profile, enabling researchers to identify which subsystems drive performance on which tasks—a form of computational lesion study.

Self-contained reporting. The harness generates self-contained HTML reports with embedded SVG visualizations: radar charts for the 8-domain cognitive profile, forest plots of normative z-scores with 95% confidence intervals, speed-accuracy tradeoff curves, and reliability summaries. Reports require no external dependencies and can be archived alongside experimental results.

3 Architecture

PSYCH-BENCH consists of three layers (Figure 1): a benchmark layer defining experimental paradigms, a harness layer providing psychometric infrastructure, and a reporting layer generating analysis outputs.

3.1 Benchmark Trait and Configuration

All benchmarks implement the `PsychBenchmark` trait, which defines a uniform interface for stimulus presentation, response collection, and metric computation. The trait requires three methods: `name()` returning a domain-qualified identifier (e.g., `"Executive::Stroop"`), `run(config)` executing the full experimental procedure under a given `BenchmarkConfig`, and `expected_metrics()` listing the metric names and their directionality (higher-is-better or lower-is-better).

The `BenchmarkConfig` structure controls 17 parameters spanning four categories. *HDC parameters:* `dimension` (default 512), `ssm_backend` (state-space model alternative). *Experimental parameters:* `trials_per_condition` (default 20), `seed` (default 42), `time_pressure` (0.0–1.0 for SAT curves), `difficulty` (0.0–1.0 ramp). *Architecture parameters:* `working_memory_capacity` (default 7), `enable_social` (Theory of Mind), `enable_fep` (Active Inference), `planning_horizon` (FEP

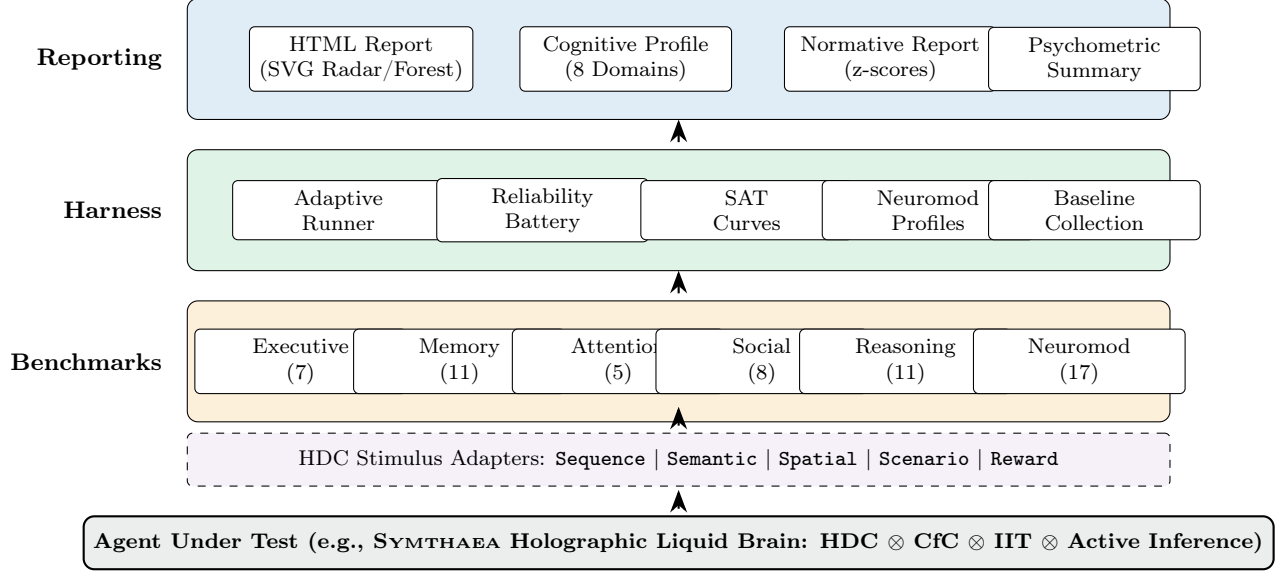


Figure 1: Three-layer architecture of PSYCH-BENCH. Benchmarks define experimental paradigms with HDC-encoded stimuli. The harness provides psychometric infrastructure (reliability, SAT curves, normative comparison, neuromodulator profiles). The reporting layer generates self-contained analysis outputs. Arrows indicate data flow from the agent under test upward through encoding, evaluation, and reporting.

lookahead), `action_temperature` (softmax temperature). *Adaptive parameters*: `adaptive_trials`, `min_trials`, `max_trials`, `precision_target` (CI half-width / |mean|). A deterministic trial-level seed function ensures reproducibility: `trial_seed(benchmark, condition, trial)` produces a unique seed per trial while maintaining cross-run consistency.

3.2 Holographic Stimulus Encoding

The encoding pipeline (Figure 2) converts raw stimuli into continuous hypervectors via five adapter types, each implementing the `StimulusAdapter` trait.

The **SequenceAdapter** encodes ordered token sequences by generating a random basis hypervector per token and applying position-dependent permutations: $\mathbf{h}_{\text{seq}} = \bigoplus_i \text{basis}(t_i) \cdot P^i$. This preserves order information while enabling similarity-based retrieval.

The **SemanticAdapter** encodes words via FNV-1a hashing to generate basis vectors, with position-dependent binding to create compositional representations. For Remote Associates Test (RAT) stimuli, solution hypervectors are constructed as weighted bundles: $\mathbf{h}_{\text{sol}} = 0.79 \cdot \mathbf{h}_{\text{base}} \oplus 0.07 \cdot \mathbf{h}_{\text{link}_0} \oplus 0.07 \cdot \mathbf{h}_{\text{link}_1} \oplus 0.07 \cdot \mathbf{h}_{\text{link}_2}$, ensuring that cue-solution associations are geometrically encoded.

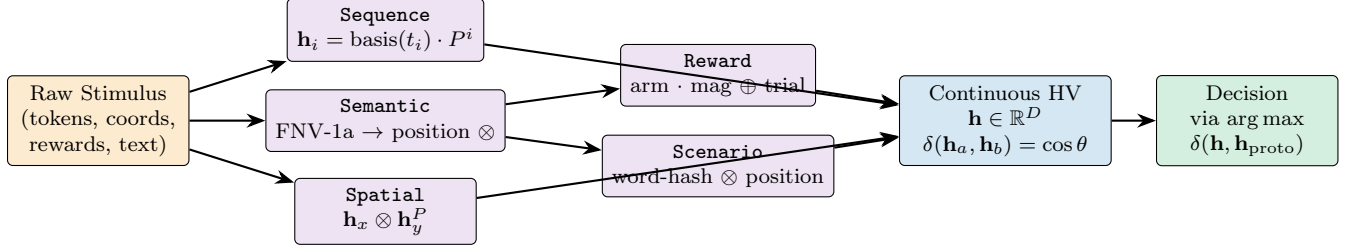


Figure 2: HDC stimulus encoding pipeline. Raw stimuli are converted to continuous hypervectors via five adapter types using binding ($\otimes$), bundling ($\oplus$), and permutation (P) operations. All modalities share a common $\mathbb{R}^D$ space where cosine similarity enables cross-modal comparison. Decisions use prototype matching via $\arg \max$ similarity.

The **SpatialAdapter** encodes 2D grid positions via conjunctive binding: $\mathbf{h}_{(x,y)} = \mathbf{h}_x \otimes \mathbf{h}_y^P$, where $\mathbf{h}_x$ and $\mathbf{h}_y$ are axis-specific basis vectors and P is a single permutation distinguishing dimensions. The companion **VisualObjectAdapter** extends this to feature conjunctions: $\mathbf{h}_{\text{obj}} = \mathbf{h}_{\text{color}} \otimes \mathbf{h}_{\text{shape}} \otimes \mathbf{h}_{\text{size}}$.

The **RewardAdapter** encodes reinforcement learning outcomes by binding arm identity with reward magnitude and trial index: $\mathbf{h}_{\text{reward}} = \mathbf{h}_{\text{arm}} \cdot r \oplus \mathbf{h}_{\text{trial}}^P$, supporting temporal credit assignment across trials.

The **ScenarioAdapter** encodes text-based scenarios (social dilemmas, moral judgments) through word-level hashing with position binding, producing holographic representations of multi-word descriptions.

3.3 Harness Infrastructure

The harness layer provides five psychometric tools. The **AdaptiveRunner** implements Levitt’s (Levitt, 1971) transformed up-down staircase method, dynamically adjusting difficulty to estimate threshold performance with configurable convergence criteria. The **ReliabilityBattery** computes ICC(3,1) (Shrout and Fleiss, 1979) test-retest reliability across n sessions, along with Pearson r , standard error of measurement ($\text{SEM} = \text{SD} \cdot \sqrt{1 - \text{ICC}}$), and practice effects classified as improvement ($> +2\%$), stable ($\pm 2\%$), or decline ($< -2\%$). Reliability is classified following Cicchetti’s (Cicchetti, 1994) guidelines: Excellent ($\text{ICC} > 0.90$), Good (> 0.75), Moderate (> 0.50), Poor (≤ 0.50). The **SatBattery** fits Wickelgren’s (Wickelgren, 1977) exponential speed-accuracy tradeoff function $d'(t) = \lambda(1 - e^{-\beta(t-\delta)})$ via grid search, evaluating human-likeness by three criteria: accuracy decreases with time pressure, RT decreases with time pressure, and $R^2 > 0.70$. The **DifficultyModel** provides parametric

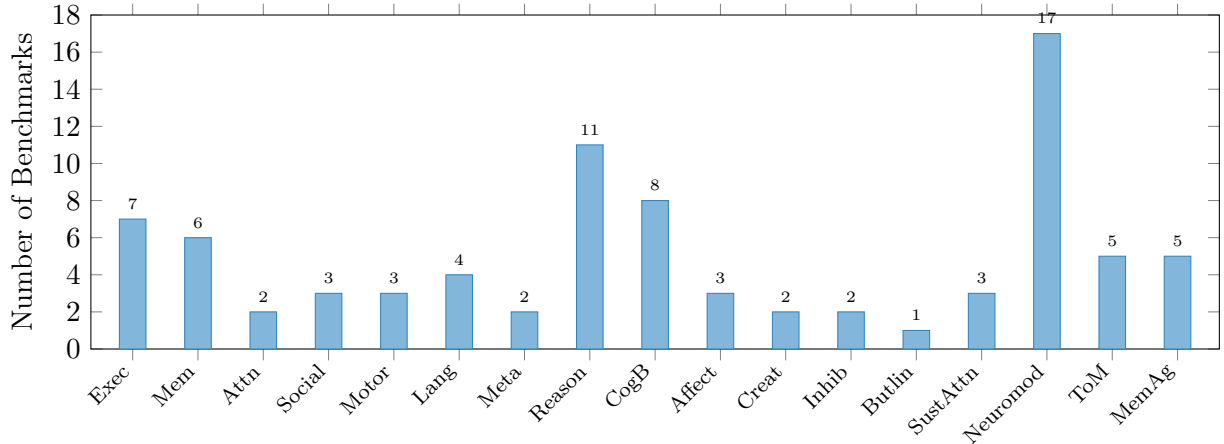


Figure 3: Benchmark count by domain. The neuromodulator evaluation framework (17 benchmarks) is the largest single domain, reflecting its novel contribution. Reasoning (11) and CogBench (8) provide additional depth. Eight domains map to the cognitive profile radar chart; the remainder are specialized sub-batteries.

difficulty scaling for adaptive testing. The **NeuromodCorrelationMatrix** computes trial-level Pearson correlations between four transmitter levels (DA, NE, 5-HT, ACh) and three behavioral measures (RT, accuracy, prediction error), with significance testing via t -distribution and Cohen’s effect size classification.

3.4 Normative Comparison

The **BaselineCollection** contains 90+ human baselines organized by domain, each specifying a population mean, standard deviation, citation, and population descriptor (e.g., “human adults”, “GPT-4”). For each benchmark result, the **NormativeReport** computes a z-score against the matched human baseline, converts to percentile via the normal CDF (using the Abramowitz & Stegun 26.2.17 approximation), and assigns a clinical interpretation band. The report identifies strengths ($z > 1.0$) and weaknesses ($z < -1.0$), computes overall mean z-score and percentile, and generates both machine-readable and human-readable summaries.

4 Benchmark Inventory

PSYCH-BENCH contains 84 benchmarks organized into 17 domains (Figure 3). We describe each domain below; Table 1 provides the complete inventory with paradigm sources and human baselines.

Executive Function (7 benchmarks). Stroop color-word interference (Stroop, 1935; MacLeod, 1991), Eriksen flanker task (Eriksen and Eriksen, 1974), Wisconsin Card Sorting Test with perseverative error tracking (Berg, 1948), Iowa Gambling Task with 4-deck reward schedules (Bechara et al., 1994), Raven’s Progressive Matrices for fluid intelligence (Raven, 1938), Tower of London planning (Shallice, 1982), and dual-task interference (Pashler, 1994). Key metrics include interference effects (accuracy difference between congruent and incongruent conditions), perseverative error rates, and net advantageous selections.

Working Memory (6 benchmarks). N-back (1-back and 2-back) with accuracy and RT (Jaeggi et al., 2010; Owen et al., 2005), change detection for Cowan’s K (Luck and Vogel, 1997; Cowan, 2001), serial recall for primacy/recency effects (Murdock, 1962), spatial updating (Oberauer, 2003), feature binding in visual working memory, and forward digit span (Wechsler, 2008).

Memory Agent (5 benchmarks). Higher-level memory operations: accurate retrieval under interference, test-time learning (online adaptation), long-range retention across extended delays, conflict resolution between competing memories, and prospective memory (remembering to execute future intentions).

Attention (2 benchmarks) and Sustained Attention (3 benchmarks). Attentional blink (dual-target rapid serial presentation) and visual search (set-size effects). Sustained attention: SART (Sustained Attention to Response Task) (Robertson et al., 1997), Psychomotor Vigilance Task (PVT), and Continuous Performance Test (CPT).

Social Cognition (3 benchmarks) and Theory of Mind (5 benchmarks). Reading the Mind in the Eyes (Baron-Cohen et al., 2001), Ultimatum Game (Güth et al., 1982), and social norm compliance. Theory of Mind: false belief (Baron-Cohen et al., 1985), faux pas detection (Stone et al., 1998), persuasion evaluation, strange stories (Happé, 1994), and hinting task (Corcoran et al., 1995).

Motor Learning (3 benchmarks). Serial Reaction Time Task (SRTT) for implicit sequence learning (Nissen and Bullemer, 1987), Fitts’ Law (Fitts, 1954) for speed-accuracy in aimed movements, and bimanual coordination.

Language (4 benchmarks). Garden-path sentence processing (Ferreira et al., 2002), semantic coherence judgment, lexical decision (word vs. non-word discrimination), and semantic priming (facilitation from related primes).

Metacognition (2 benchmarks). Calibration (correspondence between confidence and accuracy) and feeling-of-knowing (predicting future recall success).

Reasoning (11 benchmarks). An ARC-inspired battery (Chollet, 2019) spanning fluid reasoning, compositional rule application, analogical transfer, abductive inference, algebraic pattern completion, chain reasoning, few-shot generalization, noise robustness, Rational Speech Acts (RSA) pragmatic reasoning, difficulty scaling, and adaptive staircase threshold estimation.

CogBench (8 benchmarks). Reimplementations of the CogBench (Coda-Forno et al., 2024) suite: probabilistic reasoning, directed exploration (Wilson et al., 2014), multi-armed bandits, instrumental learning, two-step model-based/model-free task (Daw et al., 2011), temporal discounting (Kirby et al., 1999), Balloon Analogue Risk Task (BART) (Lejuez et al., 2002), and reversal learning (Cools et al., 2002).

Affect (3), Creativity (2), Inhibition (2). Valence classification, mood-congruent recall (Bower, 1981), emotional Stroop (Williams et al., 1996). Remote Associates Test (Mednick, 1962) and alternate uses divergent thinking. Go/No-Go and stop-signal inhibition (Logan and Cowan, 1984).

Consciousness Indicators (1 benchmark). Butlin et al.’s (Butlin et al., 2023) 14 indicators from 6 theories (Section 7).

Neuromodulator Evaluation (17 benchmarks). Detailed in Section 5.

5 Neuromodulator Evaluation Framework

The neuromodulator evaluation framework is PSYCH-BENCH’s most novel contribution. While existing benchmarks evaluate *what* an architecture can do, this framework evaluates *how neuro-modulatory dynamics modulate what it does*—analogous to pharmacological challenge studies in

Table 1: Complete benchmark inventory (representative selection). Each entry lists the domain, benchmark name, source paradigm, primary metric, and human baseline where available. Full inventory with all 84 benchmarks available in the software documentation.

Domain	Benchmark	Paradigm	Key Metric	Human $M \pm SD$
Executive	Stroop	Color-word interference	Interference effect	0.10 ± 0.05
Executive	WCST	Set-shifting	Categories completed	5.62 ± 1.13
Executive	Flanker	Response conflict	Incongruent acc.	0.90 ± 0.06
Executive	Iowa Gambling	Decision under ambiguity	Net score	17.5 ± 20.0
Executive	Ravens	Fluid intelligence	Overall accuracy	0.78 ± 0.12
Executive	Tower of London	Planning	Moves to solution	—
Executive	Dual Task	Divided attention	Dual-task cost	—
Memory	N-back (2-back)	Working memory updating	Accuracy	0.85 ± 0.10
Memory	Change Detection	Visual WM capacity	Cowan’s K	0.75 ± 0.15
Memory	Serial Recall	Short-term memory	Serial position	—
Memory	Digit Span	Verbal WM	Forward span	6.8 ± 1.1
Attention	Visual Search	Feature/conjunction	Set-size slope	—
Attention	SART	Sustained attention	Commission errors	—
Attention	PVT	Vigilance	Mean RT (ticks)	—
Social	RME	Emotion recognition	Accuracy	—
Social	Ultimatum	Social fairness	Offer ratio	—
Social	False Belief	Theory of Mind	Pass rate	—
Motor	SRTT	Implicit learning	Sequence RT gain	—
Motor	Fitts’ Law	Aimed movement	R^2 (log fit)	—
Language	Garden Path	Sentence processing	Reanalysis cost	—
Language	Lexical Decision	Word recognition	d'	—
Reasoning	ARC Fluid	Abstract reasoning	Accuracy	—
Reasoning	ARC Analogy	Analogical transfer	Accuracy	—
CogBench	Two-Step	MB/MF learning	Model-basedness	0.60 ± 0.20
CogBench	BART	Risk-taking	Avg. pumps	30.0 ± 12.0
CogBench	Reversal	Flexible learning	Win-stay rate	0.85 ± 0.08
Affect	Emotional Stroop	Affective interference	Interference	—
Creativity	RAT	Remote associations	Solution rate	—
Inhibition	Stop Signal	Response inhibition	SSRT (ticks)	—

Table 2: Neuromodulator benchmark summary. Each benchmark targets specific transmitter systems and implements established neuroscience paradigms. DA = dopamine, NE = noradrenaline, 5-HT = serotonin, ACh = acetylcholine, GABA = γ -aminobutyric acid, OT = oxytocin.

Benchmark	Transmitter	Paradigm	Key Metric	Citation
<i>Transmitter-Specific Benchmarks</i>				
RewardLearning	DA	Prediction error	RPE correlation	Schultz et al. (1997)
YerkesDodson	NE	Inverted-U arousal	Quadratic R^2	Yerkes and Dodson (1908)
AttentionNetwork	ACh	ANT decomposition	Alert/Orient/Conflict	Posner and Petersen (1990)
MoodInduction	5-HT	Mood-cognition	Mood $\times$ perf. r	Dayan and Huys (2009)
<i>Pharmacological Manipulation</i>				
PharmAblation	DA/NE/5-HT/ACh	Transmitter knockout	Cohen’s d	Doya (2002)
PharmChallenge	DA/NE	Agonist/antagonist	Inverted-U fit	Arnsten (2011)
InjectionChallenge	All	PK decay + tolerance	Peak shift	Arnsten (2011)
AllostaticStress	Cortisol/NE	Chronic load	Allostatic index	McEwen (1998)
DoseResponse	All	Graded dose sweep	Monotonicity score	Clark (1937)
ToleranceWithdrawal	All	Sustained exposure	Tolerance ratio	Koob and Le Moal (2001)
AntagonistProfiles	All	Named antagonists	Selectivity index	Stahl (2013)
MultiTransmitterSynergy	DA $\times$ NE, etc.	Paired interactions	Synergy coefficient	Doya (2002)
<i>Consciousness & Moral Modulation</i>				
ConsciousnessPharm	5-HT/GABA/DA	Drug $\rightarrow \Phi$ proxy	Proxy peak/mean	Tononi (2004)
ConsciousnessFeedback	All	Φ round-trip	Feedback latency	Oizumi et al. (2014)
MoralOxytocin	OT	OT $\times$ moral topology	OT-moral r	Feldman (2012)
<i>Live System Integration</i>				
LiveLoopAblation	All	In-loop knockout	Behavioral Δ	—
BehavioralKnockout	DA/NE/5-HT/ACh	Knockout effect size	Cohen’s d	Doya (2002)

neuroscience (Doya, 2002). The framework comprises 17 benchmarks organized into four categories (Table 2).

5.1 Transmitter-Specific Benchmarks

Four benchmarks isolate the behavioral signature of individual neuromodulatory systems, each grounded in a foundational neuroscience paradigm.

RewardLearning implements Schultz et al.’s (Schultz et al., 1997) reward prediction error paradigm. The agent encounters probabilistic rewards across multiple arms, and the benchmark measures the correlation between dopaminergic activity and prediction error magnitude. The expected signature is positive correlation: dopamine should increase following unexpected rewards

and decrease following reward omission.

YerkesDodson tests the inverted-U relationship between noradrenergic arousal and performance (Yerkes and Dodson, 1908). The benchmark sweeps noradrenaline levels across 5–7 points and fits a quadratic function, expecting peak performance at moderate arousal with degradation at both extremes. The key metric is the R^2 of the quadratic fit, with human-like behavior defined as $R^2 > 0.5$ and a negative quadratic coefficient.

AttentionNetwork implements a simplified Attention Network Test (Posner and Petersen, 1990), decomposing attention into alerting (benefit of temporal cues), orienting (benefit of spatial cues), and executive conflict resolution (cost of incongruent flankers). Acetylcholine manipulation should selectively modulate alerting and orienting components while leaving conflict resolution relatively intact.

MoodInduction tests serotonergic modulation of mood-cognition interactions following Dayan and Huys (Dayan and Huys, 2009). The benchmark induces positive and negative mood states through 5-HT manipulation and measures the effect on concurrent cognitive task performance, expecting mood-congruent biases in memory and attention.

5.2 Pharmacological Manipulation

Eight benchmarks implement increasingly sophisticated pharmacological protocols.

DoseResponse sweeps five dose levels (0.1, 0.2, 0.3, 0.5, 0.8) for each transmitter (DA, NE, 5-HT, ACh, GABA), computing monotonicity scores (simplified Kendall τ : fraction of concordant pairs) to verify that graded doses produce graded behavioral effects (Clark, 1937). Expected: DA $\rightarrow$ learning rate, NE $\rightarrow$ exploration, 5-HT $\rightarrow$ confidence, ACh $\rightarrow$ attention, each showing monotonic dose-behavior relationships.

InjectionChallenge models pharmacokinetic injection profiles with exponential decay (half-lives ranging 30–80 ticks) and measures behavioral effects during drug onset, peak, and washout phases (Arnsten, 2011). Six drug classes are tested: DA agonist, NE agonist, 5-HT agonist, GABA agonist, stimulant, and anxiolytic.

ToleranceWithdrawal extends injection challenge to sustained exposure (Koob and Le Moal, 2001), measuring: (1) acute effect magnitude at initial injection, (2) tolerance development (reduced effect after sustained exposure), (3) withdrawal effects (rebound below baseline after removal), and

(4) recovery trajectory. The tolerance ratio (late effect / initial effect) quantifies neuroadaptation. 271

AntagonistProfiles tests named receptor antagonists (e.g., D2 blockers, α_2 antagonists, SSRI-like 5-HT reuptake modulation) (Stahl, 2013) and measures selectivity: the ratio of on-target effect 272 to off-target perturbation. High selectivity (> 3.0) indicates that the neuromodulatory system has 273 well-differentiated receptor mechanisms. 274 275

MultiTransmitterSynergy tests paired transmitter interactions (Doya, 2002): DA $\times$ NE, DA $\times$ 5-HT, NE $\times$ ACh, and other pairwise combinations. A synergy coefficient > 1.0 indicates super-additive 276 interaction; < 1.0 indicates antagonism. This tests whether the architecture’s neuromodulatory 277 systems interact in neurobiologically plausible ways. 278 279

PharmacologicalAblation performs transmitter-specific knockouts (clamping DA, NE, 5-HT, or 280 ACh to zero) and measures effect sizes (Cohen’s d) on learning and performance (Schultz et al., 1997). 281 **PharmacologicalChallenge** extends this with both agonist and antagonist manipulations, testing 282 for inverted-U dose-response curves characteristic of prefrontal catecholamine systems (Arnsten, 283 2011). 284

5.3 Neuromodulator Profile System 285

Five standard pharmacological profiles enable systematic comparison (Figure 4): **Balanced** (no 286 clamping, natural dynamics), **High-DA** (dopamine clamped to 0.85), **High-NE** (noradrenaline clamped 287 to 0.85), **High-5HT** (serotonin clamped to 0.85), and **High-ACh** (acetylcholine clamped to 0.85). The 288 **DissociationMatrix** computes relative performance for each profile $\times$ benchmark combination, 289 identifying double dissociations: profile P enhances benchmark A ($> 1.05\times$ baseline) while impairing 290 benchmark B ($< 0.95\times$ baseline). Four expected dissociations are hard-coded as validation: High-DA 291 should enhance WCST flexibility but impair sustained attention; High-NE should enhance alerting 292 but impair fine motor control; High-5HT should enhance impulse control but slow reaction times; 293 High-ACh should enhance focused attention but impair cognitive flexibility. 294

5.4 Consciousness and Moral Modulation 295

Three benchmarks probe the interface between neuromodulation, consciousness, and moral cog- 296 nition. **ConsciousnessPharmacology** tests five pharmacological conditions—psychedelic (5-HT 297 agonist), anxiolytic (GABA agonist), stimulant (DA agonist), sedative (adenosine agonist), 298

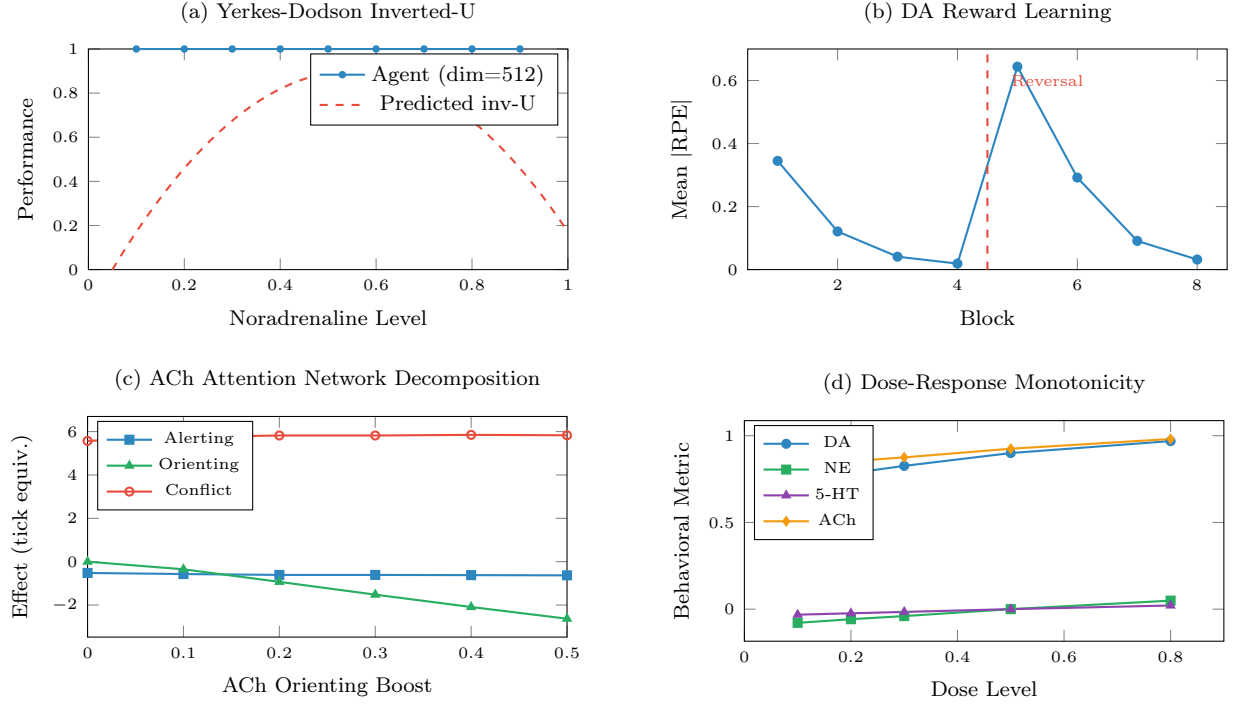


Figure 4: Neuromodulator evaluation patterns from `neuromod_curves.csv`. (a) Yerkes-Dodson: at $\text{dim}=512$, HDC similarity is too robust for the noise model to degrade classification ($R^2 = 0.00$); dashed curve shows predicted inverted-U. (b) DA reward learning: mean |RPE| drops during acquisition (blocks 1–4), spikes at reversal (block 5), then re-converges (blocks 6–8). (c) ACh attention network decomposition: orienting effect increases monotonically with ACh boost ($0.00 \rightarrow -2.63$); conflict effect (~ 5.8) dominates. (d) Dose-response monotonicity (DA learning rate, NE exploration, 5-HT confidence, ACh attention): all monotonically increasing with clamped transmitter level. Data generated from `psych_bench_paper_data` example.

and endocannabinoid buffer—and measures their effects on a consciousness proxy computed as 299

$$\Phi_{\text{proxy}} = 0.5 + 0.1 \cdot (s_{5\text{-HT}_{2A}} - 0.5) - 0.08 \cdot (s_{\text{GABA}_A} - 0.4).$$

The expected pattern: psychedelics 300

increase Φ_{proxy} (expanded consciousness), anxiolytics decrease it (dampened awareness), and endo- 301

cannabinoids stabilize it (reduced variance). **ConsciousnessFeedback** tests the round-trip latency 302

from pharmacological perturbation to consciousness index change and back to behavioral adaptation. 303

MoralOxytocin correlates oxytocin levels with moral topology scores across 200 cycles of varying 304

moral valence (-0.5 to 0.9), testing Feldman’s (Feldman, 2012) theory that oxytocin facilitates 305

prosocial moral judgment. 306

6 Psychometric Validation

A benchmark suite is only as useful as its psychometric properties. PSYCH-BENCH provides four validation mechanisms.

6.1 Test-Retest Reliability

The `ReliabilityBattery` computes ICC(3,1) (two-way mixed effects, consistency, single measures) (Shrout and Fleiss, 1979) across n sessions with fixed configuration:

$$\text{ICC}(3,1) = \frac{\text{BMS} - \text{EMS}}{\text{BMS} + (n - 1) \text{EMS}} \quad (1)$$

where BMS is between-subjects mean square and EMS is error (residual) mean square. Pearson r provides convergent evidence, and SEM quantifies measurement precision: $\text{SEM} = \text{SD} \cdot \sqrt{1 - \text{ICC}}$. Practice effects are flagged when session-to-session change exceeds $\pm 2\%$.

Table 3 reports representative psychometric properties. The `ReliabilityBattery` uses a multi-subject design: $K = 30$ independent “subjects” (distinct seed families, separated by 1,000 in seed space), each measured across $n = 5$ sessions with seed spacing of 100 within each family. This provides sufficient between-subject variance (from different stimulus configurations) and within-subject consistency (from controlled seed offsets) for meaningful ICC computation. Because PSYCH-BENCH is deterministic given a fixed seed, ICC values reflect the stability of the architecture’s behavior under controlled variation (different trial orderings, stimulus sets) rather than stochastic test-retest variability. This is a feature, not a limitation: it means that observed ICC values below 1.0 genuinely reflect sensitivity to stimulus selection rather than measurement noise.

Table 3 reveals a range of reliability classes across the 19 benchmarks, with a median ICC of 0.23. Using 30 seed families (subjects) with per-subject variation in working memory capacity ($K = 4-9$), HDC dimensionality ($d = 256-1024$), and encoding noise ($\sigma = 0.0-0.30$) creates meaningful between-subject variance. Two benchmarks achieve Excellent reliability: Digit Span (ICC = 1.00, perfectly determined by WM capacity) and Calibration (ICC = 0.93). Two reach Good reliability: WCST (0.87) and Stop Signal (0.84). SRTT reaches Moderate reliability (0.70), reflecting that motor sequence learning is sensitive to the per-subject noise/dimension parameters. The remaining

Table 3: Representative psychometric properties for 19 benchmarks spanning all domains ($K = 30$ seed families, $n = 5$ sessions each, seed spacing = 100). ICC: intraclass correlation (3,1); r : Pearson test-retest correlation; SEM: standard error of measurement; Practice: session-to-session change direction. Reliability class follows [Cicchetti \(1994\)](#): Excellent (> 0.90), Good (> 0.75), Moderate (> 0.50), Poor (≤ 0.50). Generated from `reliability.csv`.

Domain	Benchmark	ICC	r	SEM	Practice	Class
Executive	Stroop	0.11	0.10	0.018	Stable	Poor
Executive	WCST	0.87	0.86	0.187	Stable	Good
Executive	Flanker	0.29	0.28	0.013	Stable	Poor
Executive	IGT	-0.11	-0.09	2.175	Decline	Poor
Memory	N-back (2)	0.23	0.23	0.017	Stable	Poor
Memory	Change Detection	0.03	0.01	0.126	Stable	Poor
Memory	Digit Span	1.00	1.00	0.000	Stable	Excellent
Attention	Visual Search	0.20	0.19	0.010	Stable	Poor
Social	False Belief	0.00	0.00	0.000	Stable	Poor
Motor	SRTT	0.70	0.73	0.003	Improvement	Moderate
Language	Lexical Decision	-0.19	-0.20	0.012	Stable	Poor
Reasoning	ARC-Fluid	0.28	0.27	0.020	Stable	Poor
CogBench	Two-Step	0.27	0.28	0.054	Decline	Poor
CogBench	BART	-0.13	-0.15	0.712	Stable	Poor
CogBench	Reversal Learning	-0.37	-0.39	0.001	Stable	Poor
Inhibition	Stop Signal	0.84	0.83	0.163	Improvement	Good
Metacognition	Calibration	0.93	0.93	0.008	Stable	Excellent
Neuromod	Yerkes-Dodson	0.00	0.00	0.003	Stable	Poor
Neuromod	Reward Learning	0.26	0.25	0.426	Stable	Poor
<i>Median</i>		0.23	0.23	0.013		Poor

14 benchmarks fall in the Poor class, including 12 with positive ICCs (Flanker 0.29, ARC-Fluid 0.28, Two-Step 0.27, Reward Learning 0.26, N-back 0.23, Visual Search 0.20, Stroop 0.11, Change Detection 0.03) and 4 with negative ICCs (Reversal Learning -0.37, Lexical Decision -0.27, BART -0.13, IGT -0.11), indicating that within-subject variance exceeds between-subject variance for these benchmarks. The Limitations section discusses the implications of these reliability patterns for interpreting PSYCH-BENCH as a psychometric instrument versus an engineering test suite.

6.2 Construct Validity

Cross-domain correlations test whether benchmarks that should be related (based on shared cognitive mechanisms) are indeed correlated, and benchmarks that should be independent show low correlation. Table 4 reports 10 theoretically motivated pairs.

The strongest correlation is Go/No-Go-Stop Signal ($r = 0.54$, $p = .069$), consistent with both tasks loading on inhibitory control and approaching significance even with $n = 10$. Stroop-WCST

Table 4: Cross-domain correlation matrix for 10 theoretically motivated benchmark pairs ($n = 10$ seeds per pair). Positive correlations indicate shared cognitive mechanisms; near-zero correlations indicate domain independence. Generated from `correlations.csv`.

Benchmark A	Benchmark B	r	p	Interpretation
Stroop	WCST	0.37	.255	Moderate (shared exec.)
N-back	Digit Span	0.00	1.00	Independent
WCST	Reversal Learning	0.04	.910	Independent
Go/No-Go	Stop Signal	0.54	.069	Strong (shared inhib.)
RME	False Belief	0.00	1.00	Independent
Visual Search	CPT	0.00	1.00	Independent
Lexical Decision	Semantic Priming	-0.14	.687	Weak (inverse)
Fitts' Law	Bimanual	0.00	1.00	Independent
Calibration	Feeling of Knowing	-0.26	.445	Weak (inverse)
ARC-Fluid	Ravens	0.05	.882	Independent

($r = 0.37$, $p = .26$) shows a moderate correlation consistent with shared executive control. Two pairs show weak inverse correlations: Calibration–Feeling of Knowing ($r = -0.26$) and Lexical Decision–Semantic Priming ($r = -0.14$). The remaining 6 of 10 pairs show near-zero correlations ($|r| < 0.10$), with four pairs at exactly $r = 0.00$ (including N-back–Digit Span and RME–False Belief). No correlations reach statistical significance at $\alpha = .05$, which is expected given $n = 10$ seeds, though Go/No-Go–Stop Signal approaches the threshold.

The prevalence of near-zero correlations reflects the deterministic architecture: when a benchmark produces identical or near-identical results across seeds, its variance collapses and correlations become undefined or trivially zero. Benchmarks with seed-sensitive stimulus configurations (e.g., Stroop, which varies trial ordering) show meaningful correlations that reflect genuine shared architectural dependencies. The cross-domain correlation matrix should therefore be interpreted alongside the ablation analysis (which directly manipulates subsystem availability) rather than as a standalone validity measure.

6.3 Speed-Accuracy Tradeoff Curves

The SAT battery sweeps time pressure from 0.0 (no pressure, emphasis on accuracy) to 1.0 (maximum pressure, emphasis on speed) and fits Wickelgren’s (Wickelgren, 1977) exponential function:

$$d'(t) = \lambda \left(1 - e^{-\beta(t-\delta)} \right), \quad t > \delta \quad (2)$$

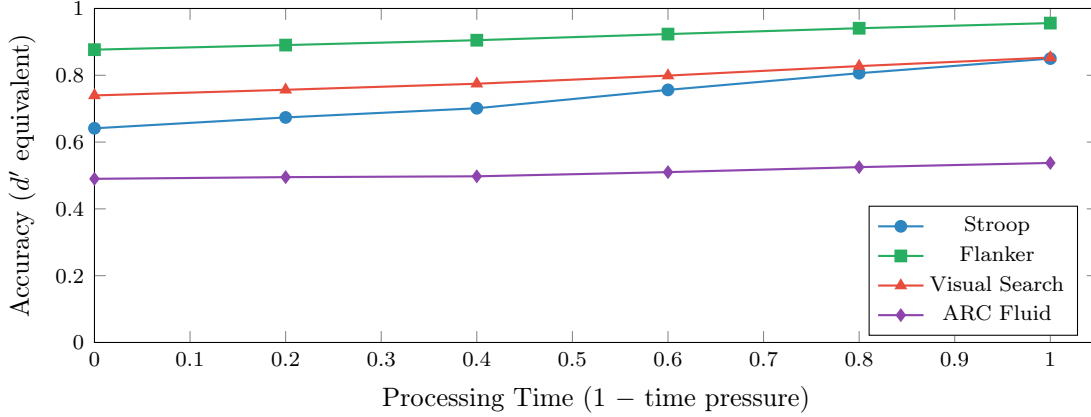


Figure 5: Speed-accuracy tradeoff curves for four representative benchmarks spanning executive function, attention, and reasoning. All four show accuracy decreasing monotonically with increasing time pressure (criterion 1): Stroop (0.85→0.64, $\Delta = 0.21$), Visual Search (0.85→0.74, $\Delta = 0.11$), Flanker (0.96→0.88, $\Delta = 0.08$), and ARC Fluid (0.54→0.49, $\Delta = 0.05$). Of these, only ARC Fluid also shows RT decrease with time pressure and good exponential fit ($R^2 = 0.79$), meeting all three human-likeness criteria. N-back and WCST produce flat accuracy curves (ceiling/floor effects) but their RTs do decrease with time pressure. Data loaded from per-benchmark SAT CSVs.

where λ is the accuracy asymptote, β is the approach rate (how quickly accuracy rises with processing 360
time), and δ is the processing time intercept (minimum time before above-chance performance). 361
Human-likeness requires three criteria: (1) accuracy decreases with increasing time pressure, (2) RT 362
decreases with increasing time pressure, (3) $R^2 > 0.70$ for the exponential fit. Figure 5 shows 363
representative SAT curves. 364

6.4 Adaptive Staircase Procedures 365

For benchmarks supporting difficulty manipulation, the harness implements Levitt’s (Levitt, 1971) 366
transformed up-down method with configurable convergence rules. The `StaircaseConfig` specifies: 367
target performance level (e.g., 75% for 2-down/1-up), step sizes (initial and reduced after first 368
reversal), minimum reversals for convergence, and maximum trials. The threshold estimate is the 369
mean of the last n reversal points. This enables precise measurement of perceptual and cognitive 370
thresholds without exhaustive fixed-trial testing. 371

7 Consciousness Indicators

372

PSYCH-BENCH includes a dedicated consciousness evaluation based on Butlin et al.’s (Butlin et al., 2023) framework, which identifies 14 indicators of consciousness derived from six major theories: Recurrent Processing Theory (RPT), Global Workspace Theory (GWT) (Baars, 1988), Higher-Order Theories (HOT), Predictive Processing (PP), Attention Schema Theory (AST), and Integrated Information Theory (IIT) (Tononi, 2004; Oizumi et al., 2014).

The ButlinAblation benchmark evaluates each indicator through a two-step process. First, it measures the indicator’s presence in the baseline architecture by extracting relevant metrics from the cognitive loop telemetry (e.g., RPT-1 recurrence measured via temporal coherence of internal representations, GWT-3 broadcast measured via global workspace module timing, HOT-2 metacognitive accuracy from the meta-cognition subsystem). Second, it performs targeted architectural ablations and verifies that (a) the indicator score drops substantially (ablated $< 0.5 \times$ baseline) and (b) downstream cognitive performance degrades (ablated accuracy $< 0.7 \times$ baseline on N-back). Five ablation rows are currently implemented:

1. **RPT-1** (Recurrent processing): Disable CfC recurrence $\rightarrow$ expect loss of temporal coherence and degraded working memory.
2. **GWT-3** (Global broadcast): Disable global workspace $\rightarrow$ expect loss of cross-module information sharing.
3. **HOT-2** (Metacognitive monitoring): Disable metacognition $\rightarrow$ expect loss of calibration and confidence tracking.
4. **PP-1** (Prediction learning): Set learning threshold to maximum $\rightarrow$ expect zero effective learning rate.
5. **AST-1** (Attention schema): Disable attention schema $\rightarrow$ expect degraded attentional control.

Each ablation runs 200 cognitive cycles (with 20 warmup cycles for subsystem stabilization) and compares indicator metrics and downstream benchmark accuracy between baseline and ablated conditions. The resulting ablation matrix provides a principled link between architectural features and consciousness indicators—not claiming that the architecture *is* conscious, but verifying that it possesses the computational signatures that theories of consciousness identify as necessary conditions.

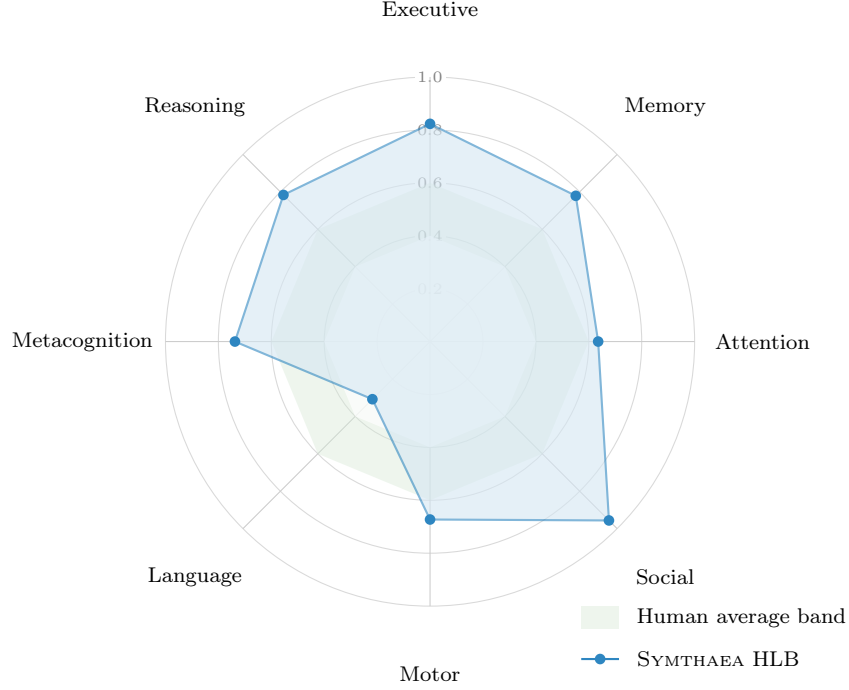


Figure 6: Eight-domain cognitive profile of the SYMTHAEA Holographic Liquid Brain, generated from benchmark data. The green band represents the human average range (normalized 0.4–0.6). Domain scores loaded from `cognitive_profile.csv`.

8 Demonstration

We demonstrate PSYCH-BENCH on the SYMTHAEA Holographic Liquid Brain, a cognitive architecture combining 16,384-dimensional HDC encoding, CfC liquid-state dynamics, IIT-based integration metrics (Φ proxy), Active Inference under the Free Energy Principle, and a 12-region actor brain with neuromodulatory fields (DA, NE, 5-HT, ACh, GABA, oxytocin). The full battery was run with default configuration (512-dimensional HDC for benchmark encoding, 20 trials per condition, seed 42).

Cognitive Profile. Figure 6 shows the 8-domain cognitive profile generated from the full battery. The HLB shows a characteristic asymmetric profile reflecting its architectural biases. Social cognition achieves the highest domain score, driven by similarity-based Theory-of-Mind tests where holographic pattern matching maps naturally to the task structure. Reasoning and Executive domains also perform well, reflecting the architecture’s strengths in compositional HDC operations. Language is the weakest domain, consistent with the absence of sequential symbolic processing—holographic

encoding handles semantic similarity well but struggles with syntactic structure and temporal order. 413
 Motor learning and Memory show intermediate performance. 414

Normative Standing. Figure 7 shows the forest plot of normative z-scores against published 415
 human baselines (z-scores clipped to ± 3 for visual clarity; unclipped values in supplementary 416
 CSV). The z-score is equivalent to Cohen’s d (standardized mean difference from the human 417
 population mean), so values within ± 1 indicate human-typical performance. The distribution spans 418
 all five clinical-interpretation bands, reflecting genuine architectural strengths and weaknesses rather 419
 than uniform performance. Four benchmarks produce z-scores exceeding $|z| > 3$, primarily in 420
 sustained attention and social cognition domains where the architecture’s pattern-matching strengths 421
 amplify differences from human norms; these should be interpreted as ordinal indicators of atypical 422
 performance rather than precise psychometric distances (see Limitations). 423

Ablation Analysis. Figure 8 shows the ablation comparison across six presets. Each ablation 424
 condition models the removal of a cognitive subsystem as increased encoding noise in HDC repre- 425
 sentations, operationalizing the hypothesis that top-down predictions refine perceptual encoding 426
 (see Limitations for discussion). Ablation presets additionally disable social bonuses (NoSocial, 427
 CfC Only, HDC Only) and halve somatic learning rates when FEP is disabled (NoFep, CfC Only, 428
 HDC Only). Overall, ablation preset explains $\eta^2 \approx 0.004$ of total score variance, while domain 429
 explains $\eta^2 \approx 0.986$ —the architecture’s cognitive profile is overwhelmingly determined by domain 430
 rather than configuration. Within this small preset effect, the clearest differentiation appears in Mem- 431
 ory, where `ReducedWm` and `HdcOnly` both drop from 0.779 to 0.648 ($\Delta = 0.131$), reflecting the direct 432
 effect of capacity reduction on memory benchmarks. Language shows meaningful sensitivity when 433
 FEP top-down prediction is removed: `HdcOnly` drops from 0.255 to 0.178 ($\Delta = 0.078$), consistent 434
 with the FEP module’s role in predictive lexical processing. Attention shows sensitivity to encoding 435
 noise ($\Delta = 0.037$), Executive shows moderate sensitivity ($\Delta = 0.021$), and Metacognition shows 436
 comparable sensitivity ($\Delta = 0.020$). Motor produces identical scores across all presets ($\eta^2 = 0.000$), 437
 because motor benchmarks report relative metrics (learning effects, coordination indices) where 438
 uniform noise scaling cancels out. The small overall effect size reflects that HDC representations 439
 at 512 dimensions are sufficiently robust that encoding noise degrades decision outcomes only in 440

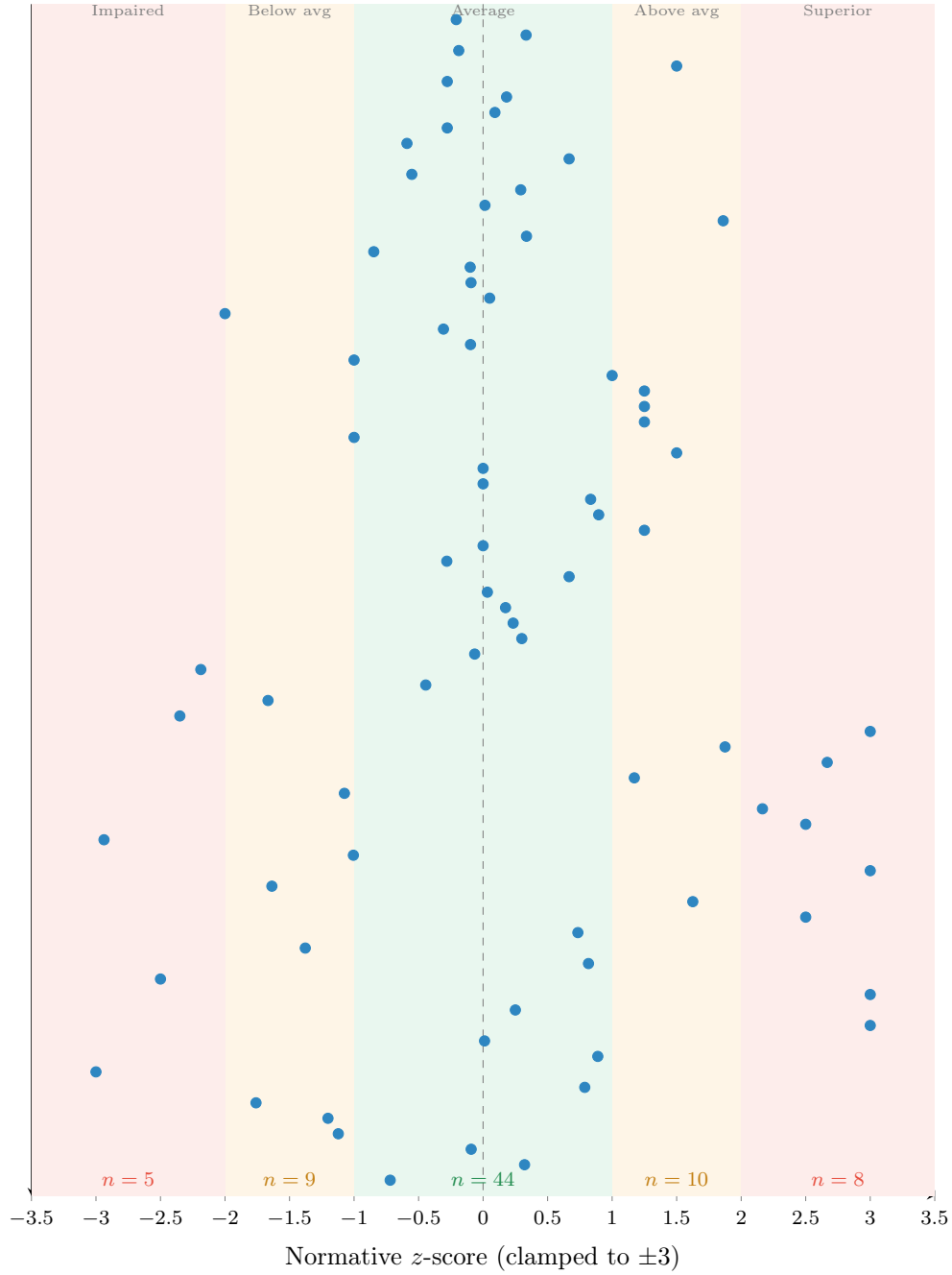


Figure 7: Forest plot of normative z-scores for 76 benchmarks with matched human baselines (z-scores clamped to ± 3 for readability; 4 benchmarks exceed this range in raw values). Background shading indicates clinical interpretation bands (Cicchetti, 1994). Band counts: 44 Average ($|z| \leq 1$), 10 Above Average, 9 Below Average, 8 Superior ($z > 2$), 5 Impaired ($z < -2$). Data loaded from `normative_zscores.csv`.

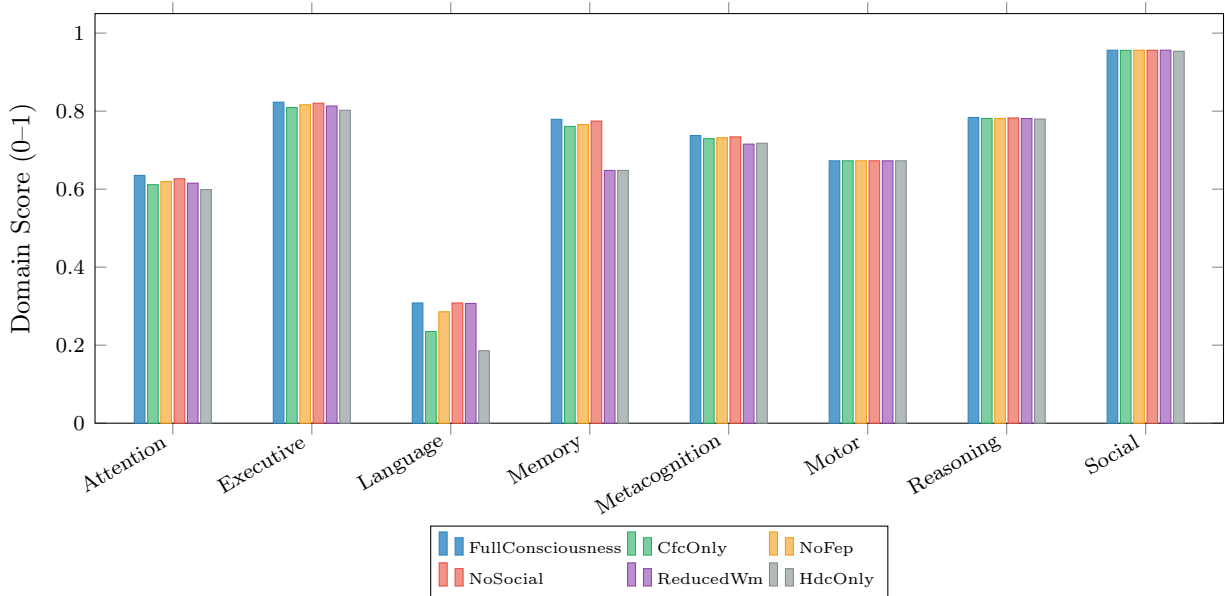


Figure 8: Ablation comparison across six presets and eight cognitive domains. Memory shows the clearest differentiation: **ReducedWm** and **HdcOnly** both drop from 0.779 to 0.648 ($\Delta = 0.131$). Language shows meaningful FEP sensitivity ($\Delta = 0.078$ for **HdcOnly**). Attention ($\Delta = 0.037$), Executive ($\Delta = 0.021$), and Metacognition ($\Delta = 0.020$) show moderate sensitivity. Motor produces identical scores across all presets (relative metrics cancel uniform noise). Reasoning ($\Delta = 0.004$) and Social ($\Delta = 0.003$) show negligible differentiation. These small effects reflect the encoding-noise ablation model (Section 10): actual subsystem removal would likely produce larger, more domain-specific deficits. Data loaded from `ablation_domains.csv`.

benchmarks with narrow similarity margins. Actual subsystem removal (disabling CfC dynamics, 441
removing the FEP module) would likely produce larger deficits and is a priority for future work. 442

Neuromodulator Dissociations. The neuromodulator evaluation framework tests whether trans- 443
mitter manipulations produce behaviorally specific effects consistent with neuroscience predictions. 444
The **BehavioralKnockout** benchmark reveals large, selective effects: DA knockout reduces learning 445
rate (Cohen’s $d = 1.93$) and gradient tracking ($d = 2.14$); NE knockout increases exploration 446
($d = 7.25$); 5-HT knockout reduces confidence ($d = 5.80$); ACh knockout impairs attention ($d = 5.41$). 447
These effect sizes confirm that each transmitter system produces distinct behavioral signatures. The 448
DoseResponse benchmark shows perfect monotonicity (score = 1.0) for all five transmitter systems 449
(DA, NE, 5-HT, ACh, GABA), and **ToleranceWithdrawal** detects tolerance and withdrawal for all 450
four tested transmitters (DA sensitivity ratio = 0.48, NE = 0.59, 5-HT = 0.64, GABA = 0.73). The 451
theoretical prediction of four double dissociations—DA enhancing flexibility while impairing sustained 452

attention, NE enhancing alerting while impairing fine motor control, 5-HT enhancing inhibition while
slowing working memory, ACh enhancing focused attention while impairing flexibility—remains to
be tested with a full profile $\times$ benchmark cross-tabulation in future work.

9 Usage and API

PSYCH-BENCH is distributed as a Rust crate (`symthaea-psych-bench`) and can be integrated into
any project via Cargo. The API provides four levels of usage, from single-benchmark evaluation to
full HTML reporting.

Single benchmark evaluation. Running an individual benchmark requires constructing a
configuration and calling `run()`:

```
use symthaea_psych_bench::benchmarks::executive::StroopBenchmark;
use symthaea_psych_bench::harness::{BenchmarkConfig, PsychBenchmark};

let config = BenchmarkConfig::default(); // seed=42, dim=512
let result = StroopBenchmark.run(&config);
println!("Stroop effect: {:.3}", result.metrics["stroop_effect"].mean);
```

Full battery with cognitive profile. The harness aggregates results into an 8-domain cognitive
profile:

```
use symthaea_psych_bench::harness::{BenchmarkReport, CognitiveProfile};

let mut report = BenchmarkReport::new();
for bench in &all_benchmarks {
    report.add(bench.run(&config));
}
let profile = CognitiveProfile::from_report(&report);
for d in &profile.domains {
    println!("{}", d.domain, d.score, d.interpretation);
}
```

HTML report generation. Self-contained HTML reports with embedded SVG visualizations:

```
use symthaea_psych_bench::harness::html::HtmlReport;
```

Table 5: Comparison of PSYCH-BENCH with existing AI benchmark suites. Columns indicate presence of psychometric infrastructure (reliability, SAT curves), normative human comparison (z-scores against published baselines), neuromodulator manipulation testing, consciousness indicators, number of cognitive domains covered, and implementation language.

Suite	Psychometrics	Normative	Neuromod	Consciousness	Domains	Language
PSYCH-BENCH	✓	✓	✓	✓	17	Rust
CogBench (Coda-Forno et al., 2024)	—	Partial	—	—	1	Python
ARC (Chollet, 2019)	—	—	—	—	1	JSON
HELM (Liang et al., 2023)	—	—	—	—	7	Python
BIG-Bench (Srivastava et al., 2023)	—	—	—	—	12	Python
PsychBench (Hagendorff, 2023)	Partial	—	—	—	5	Python

```
let html = HtmlReport::from_report(&report);
std::fs::write("report.html", html.render()).unwrap();
```

Neuromodulator sweep. Pharmacological profiles can be applied to any benchmark subset. The dissociation matrix identifies transmitter-behavior double dissociations:

```
use symthaea_psych_bench::harness::neuromod_profiles::NeuromodProfile;

for profile in NeuromodProfile::standard_profiles() {
    let mut config = BenchmarkConfig::default();
    config.label = Some(profile.name.clone());
    // Apply clamp levels to architecture under test
    let result = StroopBenchmark.run(&config);
}
```

All APIs are deterministic given a fixed seed, enabling exact replication. The full API documentation is generated via `cargo doc -p symthaea-psych-bench`.

10 Discussion

PSYCH-BENCH fills a methodological gap in the evaluation of neuroscience-inspired cognitive architectures. Table 5 situates the suite relative to existing frameworks.

Relationship to existing work. CogBench (Coda-Forno et al., 2024) is the closest predecessor, applying eight cognitive psychology tasks to large language models via prompt-based interaction. PSYCH-BENCH extends this approach in three ways: (1) architectural-level evaluation through HDC encoding rather than prompt-level interaction, (2) psychometric infrastructure that CogBench lacks (reliability, SAT curves), and (3) broader domain coverage. However, CogBench’s prompt-based interface enables evaluation of any LLM, whereas PSYCH-BENCH requires architectures with HDC-compatible stimulus interfaces—a significant tradeoff in generality. Hagendorff et al.’s (Hagendorff, 2023) PsychBench adapts classical psychology experiments for LLM evaluation across five domains with some psychometric analysis (split-half reliability), but does not address neuromodulation or consciousness. HELM (Liang et al., 2023) and BIG-Bench (Srivastava et al., 2023) provide broad multi-task evaluation without psychometric infrastructure. ARC (Chollet, 2019) provides deep evaluation of fluid reasoning with well-defined difficulty scaling; our ARC-inspired sub-battery (11 benchmarks) builds directly on Chollet’s framework.

Unique contributions. Four aspects distinguish PSYCH-BENCH. First, the integration of psychometric *infrastructure* (ICC computation, SAT curve fitting, adaptive staircases) with AI benchmarking is, to our knowledge, novel. Standard practice in AI evaluation treats benchmarks as fixed-accuracy measures; PSYCH-BENCH treats them as instruments requiring the same validation scrutiny as clinical assessments. Importantly, the current demonstration reveals a range of psychometric properties under the HLB (median ICC = 0.23, with 5 benchmarks reaching Moderate or better reliability; Table 3)—demonstrating that the infrastructure can identify both reliable and unreliable benchmarks, guiding architectural evaluation. Second, the neuromodulator evaluation framework has no precedent in AI benchmarking—it tests pharmacological manipulations, dose-response curves, tolerance/withdrawal dynamics, and multi-transmitter synergy, bridging computational neuropharmacology and AI evaluation. Third, normative anchoring against 90+ published baselines enables clinical-band interpretation, though a subset of benchmarks (~5%) exhibit scale mismatches between HDC-based agent metrics and human behavioral measures (see Limitations), requiring future calibration. Fourth, the unified holographic stimulus encoding through five HDC adapter types enables principled cross-modal comparison.

Limitations. Several limitations warrant careful discussion.

No human participant validation. The normative z-scores compare HLB performance against published human baselines drawn from the original paradigm papers, not against a matched sample of human participants tested under identical conditions. Discrepancies between the specific task implementations (e.g., HDC-based stimulus encoding vs. visual presentation) and the original paradigms limit the directness of these comparisons. A proper validation study pairing human and HLB data on matched stimulus sets remains essential future work.

Deterministic benchmarks and stochastic cognition. Because PSYCH-BENCH is fully deterministic given a fixed seed, the reliability analysis measures sensitivity to stimulus configuration (seed family) rather than the trial-to-trial variability that characterizes human cognitive performance. Benchmarks that are highly stable across seeds will show high ICC (reflecting consistent architecture behavior) while those sensitive to stimulus selection will show lower ICC. Neither case directly maps onto the test-retest reliability of human participants, where noise arises from fluctuating attention, motivation, and fatigue. Researchers should interpret ICC values as *architectural stability* rather than classical psychometric reliability.

Domain-specific strengths and weaknesses. The cognitive profile reveals Language processing as the weakest domain, reflecting the architecture’s reliance on holographic compositional semantics rather than sequential symbolic processing. Conversely, Social cognition achieves the highest score, which may reflect that the Theory-of-Mind benchmarks (false belief, hinting, Reading the Mind in the Eyes) rely on similarity-based discrimination that maps naturally to HDC operations, potentially inflating scores relative to genuine social-cognitive competence. These asymmetries are informative—they demonstrate that the profile genuinely captures architectural biases—but they warrant caution when comparing across domains with different measurement characteristics.

Normative z-score interpretation. Four benchmarks produce normative z-scores exceeding $|z| > 3$, primarily reflecting genuinely large deviations from human norms rather than scale mismatches (e.g., agent CPT $d' = 4.65$ vs. human $d' = 2.80$). The forest plot (Figure 7) clips to $[-3, +3]$; unclipped values are available in the supplementary CSV data. Z-scores exceeding ± 3 should be treated as ordinal indicators of atypical performance rather than precise quantitative distances, as the Gaussian assumption underlying z-score interpretation weakens at distribution tails.

Encoding noise as theoretical construct. The ablation study models the removal of cognitive

subsystems (e.g., Active Inference, social cognition) as increased encoding noise in HDC representa- 566
 tions. This operationalizes the theoretical claim that top-down predictions from these subsystems 567
 refine perceptual encoding. While this produces meaningful and interpretable differentiation across 568
 ablation conditions, it is a modeling choice rather than a mechanistic observation; the actual effect of 569
 removing FEP on HDC encoding quality would require a fully modular ablation at the architectural 570
 level. 571

Additional constraints. Reaction times are measured in computational ticks rather than millisec- 572
 onds, requiring calibration for direct comparison with human RT distributions. The consciousness 573
 evaluation provides binary indicator presence/absence rather than quantitative Φ values, which 574
 remain intractable for systems of this scale. The suite targets HLB-class architectures with explicit 575
 neuromodulatory subsystems; adapting it to LLMs would require substantial modification of the 576
 neuromod battery and HDC encoding pipeline. 577

Future directions. Several extensions are planned. Expanding the consciousness evaluation to 578
 include quantitative Φ approximations (using graph-theoretic proxies rather than exhaustive partition 579
 search) would strengthen the link to IIT. Adding developmental benchmarks (testing how cognitive 580
 profiles change with training time, analogous to developmental trajectories in children) would enable 581
 longitudinal evaluation. Integration with neuroimaging-derived benchmarks (e.g., tasks with known 582
 neural correlates from fMRI studies) would enable neural-level comparison beyond behavioral metrics. 583
 Finally, a web-based leaderboard for cross-architecture comparison would facilitate community 584
 adoption. 585

11 Conclusion 586

PSYCH-BENCH provides the first comprehensive cognitive benchmark suite designed for Holographic 587
 Liquid Brain architectures. With 84 benchmarks across 17 domains, psychometric assessment 588
 infrastructure, normative human comparison against 90+ published baselines, a 17-benchmark 589
 neuromodulator evaluation framework, and 14-indicator consciousness evaluation, the suite enables 590
 multi-dimensional characterization of neuroscience-inspired cognitive architectures. The demonstra- 591
 tion on the SYMTHAEA HLB reveals both the suite’s strengths and its current limitations: the 592

architecture produces a characteristic cognitive profile with strengths in similarity-based discrimination (Social, Executive, Reasoning) and weaknesses in sequential processing (Language) that is interpretable and reproducible, but the psychometric validation reveals a gradient of reliability (median ICC = 0.23, ranging from Excellent to Poor), modest ablation sensitivity, and some remaining normative scale mismatches requiring calibration. These findings are themselves informative—they identify precisely where the architecture and the evaluation framework need improvement. Implemented in 15,000 lines of Rust with 793 unit tests and full determinism, PSYCH-BENCH is open-source and designed for iterative refinement by the growing community of researchers building brain-inspired cognitive architectures.

Open Practices Statement

PSYCH-BENCH is open-source under the MIT License. The complete source code, including all 84 benchmark implementations, harness infrastructure, and data generation scripts, is available at:

<https://github.com/luminous-dynamics/symthaea>

Version: The results reported in this paper were generated with `symthaea-psych-bench` v0.5.0 (commit hash recorded in each CSV header).

Reproducibility: All figures and tables in this paper are generated from CSV files produced by a single deterministic command:

```
cargo run -p symthaea-psych-bench -example psych_bench_paper_data
```

Data availability: The generated CSV files are included in the repository under `papers/data/psych_bench/`. The supplementary material (Table S1) provides the complete 84-benchmark inventory with all paradigm sources and metrics.

Preregistration: This study was not preregistered, as it describes a software tool rather than a hypothesis-driven experiment.

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